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Section:- 23 CSE340
Assignment-1

1) By looking at opcode, we understand the instruction format and basic operation.

2) Based on funct3, machine will understand the size of data and the opcode will tell it's a load.

3a)

ld x5, 32(x20)

ld x6, 56(x21)

bne x5, x6, L1

slli x6, x6, 2

add x6, x6, x6

sd x6, 56(x21)

addi x22, x22, 10

j end

L1:

beq x5, x0, L2

slli x6, x6, 4

sd x6, 56(x21)

j end

L2:

addi x5, x5, 2

sd x5, 32(x20)

~~j end~~ addi x22, x22, 5

end

3b)

R-format:

add x6, x6, x6

00000000 00110 00110 x 00110 x

I-format:

slli x6, x6, 4

00000000 00100 00110 x 00110 x

S-format:

sd x6, 56(x21)

00000001 00110 10101 x 11000 x

4)

1) 00000000 01011 01010 000 01001 0110011
f7 rs2 rs1 f3 rd opcode

add x9, x10, x11

2) 00000000 00101 10001 101 10010 0010011
imm[11:5] imm[4:0] rs1 f3 rd opcode

slli x18, x17, 5